

Rational Equation Problem Solving Challenge

1)	$\frac{y^2 - 9y + 8}{y - 1} + \frac{3y^2 + 16y - 12}{3y - 2} = -3$	2)	Three plus the reciprocal of a number equals 7 divided by that number. What is the number?
3)	For what values of z does $\frac{4 - z}{4 + z} = \frac{z}{z + z + z + z}?$	4)	Solve the equation $\frac{3}{t - 2} + \frac{9}{2 - t} = 12$
5)	What values of x satisfy $\frac{\sqrt{x + 1} + \sqrt{x - 1}}{\sqrt{x + 1} - \sqrt{x - 1}} = 3?$	6)	Find all x , such that, $\frac{x^2 - 2x - 3}{x + 1} + \frac{x^2 + 5x - 24}{x - 3} = 13$
7)	Chewbacca has 20 pieces of cherry gum and 30 pieces of grape gum. Some of the pieces are in complete packs, while others are loose. Each complete pack has exactly x pieces of gum. If Chewbacca loses one pack of cherry gum, then the ratio of the number of pieces of cherry gum he has to the number of pieces of grape gum will be exactly the same as if he instead finds 5 packs of grape gum. Find x .	8)	If $x = \sqrt{\frac{6}{7}}$, then evaluate $\left(x + \frac{1}{x}\right)^2$.
9)	Suppose $x = z - \sqrt{z^2 - 5}$ and $5y = z + \sqrt{z^2 - 5}$, find x when $y = 2/3$	10)	Simplify: $\sqrt[3]{\frac{256}{27}} + \sqrt[3]{32} - \sqrt[3]{\frac{12}{81}}$
11)	Find the value of $\sqrt{90 + \sqrt{90 + \sqrt{90 + \dots}}}$		

1)	<i>Solution:</i> We can factor the quadratics in the numerators on the left side, and lo and behold, we can simplify the fractions: $\begin{aligned}\frac{y^2 - 9y + 8}{y - 1} + \frac{3y^2 + 16y - 12}{3y - 2} \\&= \frac{(y-1)(y-8)}{y-1} + \frac{(3y-2)(y+6)}{3y-2} \\&= y - 8 + y + 6.\end{aligned}$ So, our equation is $2y - 2 = -3$, which gives $y = \boxed{-1/2}$. (A quick check shows that this solution is not extraneous.)
2)	<i>Solution:</i> Let x be the number. Converting the words in the problem into an equation gives us $3 + \frac{1}{x} = \frac{7}{x}$. Subtracting $\frac{1}{x}$ from both sides gives $3 = \frac{6}{x}$. Multiplying both sides of this equation by x gives $3x = 6$, and dividing both sides of this equation by 3 gives $x = \boxed{2}$.
3)	<i>Solution:</i> We start by simplifying the fraction on the right side: $\frac{z}{z + z + z + z} = \frac{\cancel{z}}{4\cancel{z}} = \frac{1}{4}.$ Our equation is now $\frac{4 - z}{4 + z} = \frac{1}{4}$. Cross-multiplying gives $4(4 - z) = 1(4 + z)$. Expanding both sides gives $16 - 4z = 4 + z$. Adding $4z$ to both sides gives $16 = 4 + 5z$. Subtracting 4 from both sides results in $12 = 5z$. Finally, we divide both sides by 5 to find $z = \boxed{12/5}$. This value of z does not make any of the denominators 0 in the original equation, so it is a valid solution.
4)	<i>Solution:</i> At first it might look like we cannot easily combine the two terms on the left side because they have different denominators. However, the second denominator is just the negative of the first: $-(t - 2) = -t - (-2) = -t + 2 = 2 - t.$ Therefore, we can make the two denominators the same by multiplying the numerator and denominator of the second fraction by -1. This allows us to add the fractions easily: $\begin{aligned}\frac{3}{t - 2} + \frac{9}{2 - t} &= \frac{3}{t - 2} + \frac{9}{2 - t} \cdot \frac{-1}{-1} \\&= \frac{3}{t - 2} + \frac{-9}{t - 2} \\&= \frac{3 - 9}{t - 2} \\&= \frac{-6}{t - 2}.\end{aligned}$ Now our equation is $\frac{-6}{t - 2} = 12$. Multiplying both sides by $t - 2$ gets rid of the fraction and leaves us $-6 = 12(t - 2)$. We might expand the right side, or first divide by 6 to give $-1 = 2(t - 2)$. Expanding the right side gives $-1 = 2t - 4$. Adding 4 to both sides of this gives $3 = 2t$, and dividing both sides of this equation by 2 gives us the solution $t = \boxed{3/2}$. This value of t does not make any of our denominators equal to 0, so it is a valid solution.

5)

Solution: We start by getting rid of the nasty denominator. We can do so by multiplying both sides by $\sqrt{x+1} - \sqrt{x-1}$. On the left, we have

$$\begin{aligned} & (\sqrt{x+1} - \sqrt{x-1}) \cdot \frac{\sqrt{x+1} + \sqrt{x-1}}{\sqrt{x+1} - \sqrt{x-1}} \\ &= \frac{(\cancel{\sqrt{x+1}} - \cancel{\sqrt{x-1}})(\sqrt{x+1} + \sqrt{x-1})}{\cancel{\sqrt{x+1}} - \cancel{\sqrt{x-1}}} \\ &= \sqrt{x+1} + \sqrt{x-1}. \end{aligned}$$

On the right, we have

$$3(\sqrt{x+1} - \sqrt{x-1}) = 3\sqrt{x+1} - 3\sqrt{x-1},$$

so our equation is

$$\sqrt{x+1} + \sqrt{x-1} = 3\sqrt{x+1} - 3\sqrt{x-1}.$$

We have two $\sqrt{x+1}$ terms, so we can combine them by subtracting $\sqrt{x+1}$ from both sides. This will leave $\sqrt{x-1}$ on the left, and on the right we'll have

$$\begin{aligned} 3\sqrt{x+1} - 3\sqrt{x-1} - \sqrt{x+1} &= (3\sqrt{x+1} - \sqrt{x+1}) - 3\sqrt{x-1} \\ &= (3-1)\sqrt{x+1} - 3\sqrt{x-1} \\ &= 2\sqrt{x+1} - 3\sqrt{x-1}. \end{aligned}$$

Now we have the equation $\sqrt{x-1} = 2\sqrt{x+1} - 3\sqrt{x-1}$. We can combine the two $\sqrt{x-1}$ terms by adding $3\sqrt{x-1}$ to both sides, which gives us $4\sqrt{x-1} = 2\sqrt{x+1}$. We can simplify this equation a little by dividing both sides by 2, which gives $2\sqrt{x-1} = \sqrt{x+1}$. Now our main obstacles are those square root signs. We can get rid of both of them by squaring both sides of the equation. On the left we have

$$(2\sqrt{x-1})^2 = 2^2(\sqrt{x-1})^2 = 4(x-1) = 4x-4.$$

On the right, squaring $\sqrt{x+1}$ gives us $x+1$, so our equation now is just $4x-4 = x+1$. Solving this equation, we find $x = \boxed{5/3}$. Make sure you plug this solution back in and see that it isn't extraneous!

6)

Solution for Problem 10.26: We could multiply both sides by $x+1$ and by $x-3$ to get rid of the fractions, but that looks pretty complicated. Before we do that, we look for another approach. The quadratics in the numerators are pretty simple; maybe we can factor them. We find

$$\begin{aligned} x^2 - 2x - 3 &= (x-3)(x+1) \quad \text{and} \\ x^2 + 5x - 24 &= (x-3)(x+8). \end{aligned}$$

Aha! In each fraction, a factor in the numerator cancels with the denominator:

$$\frac{(x-3)\cancel{(x+1)}}{\cancel{x+1}} + \frac{(\cancel{x-3})(x+8)}{\cancel{x-3}} = 13.$$

This makes our equation $(x-3) + (x+8) = 13$. Solving this equation, we find $x = 4$. This solution doesn't make any of our denominators 0, so it is indeed a valid solution. $\square$

7)

Solution: If Chewbacca loses one pack of cherry gum, the ratio of the number of pieces of cherry gum he has to the number of pieces of grape gum is $(20-x)/30$. If he instead finds 5 packs of grape gum, this ratio will be $20/(30+5x)$. These ratios must be equal, so we must have

$$\begin{aligned} \frac{20-x}{30} &= \frac{20}{30+5x} \\ \Rightarrow (20-x)(30+5x) &= (30)(20) \\ \Rightarrow (20-x)(5)(6+x) &= (30)(20). \end{aligned}$$

Dividing both sides by 5 gives $(20-x)(6+x) = (30)(4)$ and expanding the left side of this gives $120 + 14x - x^2 = 120$. Therefore, $x^2 - 14x = 0$, so $x(x-14) = 0$. We can't have $x = 0$, so we must have $x = \boxed{14}$.

8)

Solution for Problem 11.4: We could start by sticking our value for x into the expression:

$$\left(\sqrt{\frac{6}{7}} + \frac{1}{\sqrt{\frac{6}{7}}}\right)^2 = \left(\sqrt{\frac{6}{7}} + \sqrt{\frac{7}{6}}\right)^2.$$

Yuck. We could keep going and find a common denominator, then square everything, but instead we recall our expansion for the square of a binomial:

$$(a + b)^2 = a^2 + 2ab + b^2.$$

Here, $a = x$ and $b = \frac{1}{x}$, so we have

$$\left(x + \frac{1}{x}\right)^2 = x^2 + 2(x)\left(\frac{1}{x}\right) + \left(\frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}.$$

Success! Because $x^2 = \frac{6}{7}$, this expansion allows us to avoid the radicals entirely:

$$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2} = \frac{6}{7} + 2 + \frac{7}{6} = \frac{169}{42}.$$

9)

We know that squaring gets rid of square roots, but squaring either of our equations doesn't look too helpful. However, we notice that the right sides of our equations are of the form $a - b$ and $a + b$, where $a = z$ and $b = \sqrt{z^2 - 5}$. We could add them to get rid of $\sqrt{z^2 - 5}$, but that would give $x + 5y = 2z$, and we're back to having to find z .

Instead of getting rid of b , we reach back to our squaring idea to get rid of the square roots. Because $(a - b)(a + b) = a^2 - b^2$, multiplying the two equations will get rid of the square roots:

$$(x)(5y) = (z - \sqrt{z^2 - 5})(z + \sqrt{z^2 - 5}).$$

^p The right side is the factorization of a difference of squares, so we have

$$5xy = (z)^2 - (\sqrt{z^2 - 5})^2 = z^2 - (z^2 - 5) = 5.$$

Therefore, $xy = 1$. Since $y = 2/3$, we conclude that $x = 1/y = 3/2$. $\square$

10)

Solution: First, we simplify all three cube roots (and simplify $12/81$ to $4/27$):

$$\begin{aligned}\sqrt[3]{\frac{256}{27}} &= \frac{\sqrt[3]{256}}{\sqrt[3]{27}} = \frac{\sqrt[3]{2^8}}{3} = \frac{\sqrt[3]{2^3} \times \sqrt[3]{2^3} \times \sqrt[3]{2^2}}{3} = \frac{4\sqrt[3]{4}}{3}, \\ \sqrt[3]{32} &= \sqrt[3]{2^3} \times \sqrt[3]{2^2} = 2\sqrt[3]{4}, \text{ and} \\ \sqrt[3]{\frac{12}{81}} &= \sqrt[3]{\frac{4}{27}} = \frac{\sqrt[3]{4}}{\sqrt[3]{27}} = \frac{\sqrt[3]{4}}{3}.\end{aligned}$$

We then have

$$\begin{aligned}\sqrt[3]{\frac{256}{27}} + \sqrt[3]{32} - \sqrt[3]{\frac{12}{81}} &= \frac{4\sqrt[3]{4}}{3} + 2\sqrt[3]{4} - \frac{\sqrt[3]{4}}{3} \\ &= \left(\frac{4}{3} + 2 - \frac{1}{3}\right)\sqrt[3]{4} \\ &= \boxed{3\sqrt[3]{4}}.\end{aligned}$$

11)

Solution: We let the expression equal x :

$$x = \sqrt{90 + \sqrt{90 + \sqrt{90 + \cdots}}}$$

We square both sides to get rid of one radical:

$$x^2 = 90 + \sqrt{90 + \sqrt{90 + \cdots}}$$

We still have that big nasty infinite radical. However, we already said the nasty radical is equal to x ! So, we can substitute x in place of the radical to get $x^2 = 90 + x$. So, $x^2 - x - 90 = 0$, and factoring the left side gives us $(x - 10)(x + 9) = 0$. Since x is clearly positive, we have $x = \boxed{10}$.